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ABSENT : 11
CLASS : SIB 2G
OBJECT : MOBILE PROGRAMMING

Case Study 1: Temperature Conversion

The program converts temperatures from Celsius to Fahrenheit and Kelvin:

```
import 'package:konversi_suhu/konversi_suhu.dart';

Run | Debug
void main(List<String> arguments) {
  double c = 30;
  print('$c°C = ${celsiusKeKelvin(c)} K');
  print('$c°C = ${celsiusKeFahrenheit(c)} °F');
}
```

```
double celsiusKeKelvin(double celsius) {
  return celsius + 273.15;
}

double celsiusKeFahrenheit(double celsius) {
  return (celsius * 9/5) + 32;
}
```

Output:

This program works by setting the temperature value in Celsius as the initial data, then processing it using two conversion functions to convert it to Kelvin and Fahrenheit based on their respective formulas. After the calculation is complete, the conversion results are displayed along with the initial temperature value and the conversion results.

```
Building package executable...
Built konversi_suhu:konversi_suhu.
30.0°C = 303.15 K
30.0°C = 86.0 °F
```

Case Study 2: Simple Calculator

```
import 'package:kalkulator_sederhana/kalkulator_sederhana.dart';

Run | Debug
void main() {
  int angka1 = 10;
  int angka2 = 5;

  tampilkanHasil(angka1, angka2);
}
```

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```
void tampilkanHasil(int angka1, int angka2) {
  int penjumlahan = angka1 + angka2;
  int pengurangan = angka1 - angka2;
  int perkalian = angka1 * angka2;
  double pembagian = angka1 / angka2;
  int modulo = angka1 % angka2;

  print('$angka1 + $angka2 = $penjumlahan');
  print('$angka1 - $angka2 = $pengurangan');
  print('$angka1 * $angka2 = $perkalian');
  print('$angka1 / $angka2 = $pembagian');
  print('$angka1 % $angka2 = $modulo');
}
```

Output:

The program output displays the results of basic arithmetic operations such as addition, subtraction, multiplication, division, and remainder using two specified values. These values are processed using variables with numeric data types and arithmetic operators, and the results are displayed on the screen by inserting the variable values into the text so that the calculation information can be read clearly.

```
Building package executable...
Built kalkulator_sederhana:kalkulator_sederhana.
10 + 5 = 15
10 - 5 = 5
10 * 5 = 50
10 / 5 = 2.0
10 % 5 = 0
```

Case Study 3: User Data Validation

```
import 'package:validasi_data/validasi_data.dart';

Run | Debug
void main() {
  String username = 'budi123';
  String password = 'pass123';
  int umur = 17;

  bool daftar = canRegister(username, password);
  bool akses = canAccessAdultContent(username, password, umur);

  print('Dapat mendaftar: $daftar');
  print('Dapat mengakses konten: $akses');
}
```

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```dart
bool isUsernameValid(String username) {
  return username.length >= 6;
}

bool isPasswordValid(String password) {
  return password.length >= 6;
}

bool isAdult(int umur) {
  return umur >= 18;
}

bool canRegister(String username, String password) {
  return isUsernameValid(username) && isPasswordValid(password);
}

bool canAccessAdultContent(String username, String password, int umur) {
  return canRegister(username, password) && isAdult(umur);
}
```

Output:

This program works by setting user data in the form of a username, password, and age as initial values, which are then processed through several validation functions. Each function checks certain requirements, such as the length of the username and password and the minimum age limit, and then produces a logical value of true or false. The results of these checks are combined to determine whether the user can register and whether they are allowed to access special content. Finally, the program displays the results of the decision so that users can see the status of their data validation.

```
Building package executable...
Built validasi_data:validasi_data.
Dapat mendaftar: true
Dapat mengakses konten: false
```

Case Study 4: Discount Calculation

```dart
import 'package:perhitungan_diskon/perhitungan_diskon.dart';

Run | Debug
void main() {
  double hargaBarang = 120000;
  int jumlahBeli = 3;
  String kodeMember = 'GOLD';

  double total = hitungTotal(hargaBarang, jumlahBeli);
  double diskon = hitungTotalDiskon(total, kodeMember);
  double akhir = hitungHargaAkhir(hargaBarang, jumlahBeli, kodeMember);

  print('Total: Rp$total');
  print('Diskon: Rp$diskon');
  print('Akhir: Rp$akhir');
}
```

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```
double hitungTotal(double hargaBarang, int jumlahBeli) {
  return hargaBarang * jumlahBeli;
}

double diskonMember(String kodeMember) {
  if (kodeMember == 'GOLD') return 0.1;
  if (kodeMember == 'SILVER') return 0.05;
  return 0;
}

double diskonTambahan(double total) {
  return total > 300000 ? 0.05 : 0;
}

double hitungTotalDiskon(double total, String kodeMember) {
  return total * (diskonMember(kodeMember) + diskonTambahan(total));
}

double hitungHargaAkhir(double hargaBarang, int jumlahBeli, String kodeMember) {
  double total = hitungTotal(hargaBarang, jumlahBeli);
  return total - hitungTotalDiskon(total, kodeMember);
}
```

Output:

This program works with initial data in the form of item prices, purchase quantities, and membership codes as the basis for calculation. These values are processed through several functions to calculate the total price, determine the discount amount based on membership type and additional discounts if the total purchase exceeds a certain limit, and then calculate the final price after discounts. Once all calculations are complete, the program displays the total purchase amount, the discount amount obtained, and the final price to be paid so that the transaction results are clearly visible.

```
Building package executable...
Built perhitungan_diskon:perhitungan_diskon.
Total: Rp360000.0
Diskon: Rp54000.00000000001
Akhir: Rp306000.0
```

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Practicum Assignment

1. Create a simple Dart program to calculate BMI (Body Mass Index) based on the weight and height entered.

   This program asks users to enter their weight and height. The data entered is processed to calculate the BMI value using the formula of weight divided by height squared. Once the BMI value is obtained, the program compares the result with categories to determine whether it falls into the underweight, normal, overweight, or obese category. Finally, the program displays the BMI value and its category as the final result.

```dart
import 'dart:io';
import 'package:bmi/bmi.dart';

Run | Debug
void main() {
  // Meminta pengguna input berat badan dalam satuan kilogram
  stdout.write('Masukkan berat badan (kg): ');
  double berat = double.parse(stdin.readLineSync()!);
  // Meminta pengguna input tinggi badan dalam satuan meter
  stdout.write('Masukkan tinggi badan (meter): ');
  double tinggi = double.parse(stdin.readLineSync()!);

  // Memanggil fungsi dari lib, mengirim berat & tinggi, dan menyimpan hasil ke variabel bmi
  double bmi = hitungBMI(berat, tinggi);

  // Menampilkan hasil BMI
  print('\nBMI kamu: ${bmi.toStringAsFixed(2)}');
  // Memanggil fungsi kategori dan menampilkan hasilnya
  print('Kategori: ${kategoriBMI(bmi)}');
}
```

```dart
double hitungBMI(double berat, double tinggi) {
  return berat / (tinggi * tinggi); // Menghitung BMI memakai rumus dan mengirim hasil ke pemanggil fungsi
}

String kategoriBMI(double bmi) {
  if (bmi < 18.5) return 'Kurus'; // Jika BMI kurang dari 18.5 maka hasilnya Kurus
  if (bmi < 25) return 'Normal'; // Jika kurang dari 25 maka hasilnya Normal
  if (bmi < 30) return 'Overweight'; // Jika kurang dari 30 maka hasilnya Overweight
  return 'Obesitas'; // Jika semua kondisi tidak terpenuhi maka hasilnya Obesitas
}
```

Output:

```
Building package executable...
Built bmi:bmi.
Masukkan berat badan (kg): 45
Masukkan tinggi badan (meter): 1.61

BMI kamu: 17.36
Kategori: Kurus
```

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2. Create a currency conversion calculator with at least 3 types of currencies.

The program works by asking users to enter the amount of money and select the source currency and destination currency. The data entered is processed by first converting the monetary value to the base currency using the available exchange rate data. After that, the value is recalculated to the destination currency according to the applicable exchange rate. Finally, the program displays the conversion results along with the final currency type to the user.

```dart
Map<String, double> kurs = { // Membuat variabel bernama kurs bertipe Map dengan String nama mata uang dan double nilai kurs
  'USD': 1, // USD dijadikan mata uang dasar
  'IDR': 16800, // 1 USD = 16800 IDR
  'EUR': 1.15, // 1 EUR = 1.15 USD
};

double konversiMataUang(double jumlah, String dari, String ke) { // Menerima parameter jumlah uang, mata uang asal, mata uang tujuan
  // Menghitung dalam satuan USD dengan cara membagi jumlah yang dimasukkan dengan nilai kurs mata uang asal yang diambil dari data kurs
  double dalamUSD = jumlah / kurs[dari]!;
  return dalamUSD * kurs[ke]!; // Mengubah nilai USD ke mata uang tujuan, lalu mengembalikan hasilnya
}
```

```dart
import 'dart:io';
import 'package:konversi_uang/konversi_uang.dart';

Run | Debug
void main() {
  // Meminta pengguna untuk input jumlah uang
  stdout.write('Masukkan jumlah uang: ');
  // Membaca input pengguna, memastikan nilainya tidak kosong, mengubahnya dari teks menjadi angka desimal, lalu menyimpannya ke variabel jumlah
  double jumlah = double.parse(stdin.readLineSync()!);

  // Meminta pengguna untuk input mata uang asal
  stdout.write('Dari mata uang (USD/IDR/EUR): ');
  String dari = stdin.readLineSync()!.toUpperCase(); // Membaca input teks dan mengubah huruf menjadi kapital

  // Meminta pengguna untuk input mata uang tujuan
  stdout.write('Ke mata uang (USD/IDR/EUR): ');
  String ke = stdin.readLineSync()!.toUpperCase(); // Membaca input dan mengubah ke huruf besar

  // Memanggil fungsi dengan mengirimkan nilai jumlah, mata uang asal, dan tujuan, lalu menyimpan hasilnya ke variabel hasil
  double hasil = konversiMataUang(jumlah, dari, ke);

  // Menampilkan hasil, menyisipkan nilai variabel yang diformat menjadi dua angka desimal, lalu menambahkan kode mata uang tujuan
  print('\nHasil konversi: ${hasil.toStringAsFixed(2)} $ke');
}
```

Output:

```
Building package executable...
Built konversi_uang:konversi_uang.
Masukkan jumlah uang: 1000000
Dari mata uang (USD/IDR/EUR): idr
Ke mata uang (USD/IDR/EUR): usd

Hasil konversi: 59.52 USD
```

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3. Create a program that implements all types of operators that have been learned in a console application.

   The program works by executing operations using various types of operators. The initial values of variables are processed through arithmetic, increment decrement, relational, logical, assignment, null aware, type test, ternary, and bitwise operations to produce calculation results or condition evaluations. The program displays all of these results as a demonstration of the use of operators in a single execution process.

```dart
import 'package:semua_operator/semua_operator.dart';

Run | Debug
void main() {
  OperatorDemo().jalankan(); // Membuat objek dari class OperatorDemo, lalu memanggil method jalankan()
}
```

```dart
class OperatorDemo {
  void jalankan() {

    // Variabel awal
    int a = 10;
    int b = 3;

    // ARITMATIKA
    print("Aritmatika:");
    print(a + b); // penjumlahan
    print(a - b); // pengurangan
    print(a * b); // perkalian
    print(a / b); // pembagian
    print(a ~/ b); // pembagian bulat
    print(a % b); // sisa bagi

    // INCREMENT DECREMENT
    print("\nIncrement Decrement:");
    int x = 5; // variabel awal
    print(x++); // tampilkan nilai dulu, baru tambah 1
    print(++x); // tambah dulu, baru tampilkan
    print(x--); // tampilkan dulu, lalu kurangi
    print(--x); // kurangi dulu, baru tampilkan

    // RELASIONAL
    print("\nRelasional:");
    print(a == b); // apakah sama
    print(a != b); // apakah tidak sama
    print(a > b); // lebih besar
    print(a < b); // lebih kecil
    print(a >= b); // lebih besar atau sama
    print(a <= b); // lebih kecil atau sama
```

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```
  // LOGIKA
  print("\nLogika:");
  bool p = DateTime.now().second % 2 == 0; // p bernilai true jika detik sekarang genap
  bool q = false; // q selalu false
  print(p && q); // true jika keduanya true
  print(p || q); // true jika salah satu true
  print(!p); // kebalikan nilai p

  // ASSIGNMENT
  print("\nAssignment:");
  int n = 5; // nilai awal
  n += 3; // n = n + 3
  n -= 2; // n = n - 2
  n *= 4; // n = n × 4
  n ~/= 3; // n = n dibagi bulat 3
  n %= 2; // n = sisa bagi 2
  print(n); // menampilkan hasil akhir

  // NULL AWARE
  print("\nNull-aware:");
  String? nama; // variabel boleh null
  print(nama ?? "Default"); // jika null tampilkan Default
  nama ??= "Budi"; // isi Budi hanya jika nama null
  print(nama); // // tampilkan nilai nama
  print(nama.length); // panjang teks nama
```

```
  // TYPE TEST
  print("\nType Test:");
  dynamic data = 100; // tipe bebas
  print(data is int); // cek apakah int
  print(data is! String); // cek apakah bukan string

  dynamic teks = "halo"; // variabel teks
  String hasilCast = teks as String; // mengubah tipe menjadi string secara paksa
  print(hasilCast); // tampilkan hasil

  // TERNARY
  print("\nTernary:");
  int nilai = 75; // nilai angka
  String hasil = nilai >= 70 ? "Lulus" : "Gagal"; // jika nilai lebih besar atau sama 70 maka Lulus, selain itu Gagal
  print(hasil); // tampilkan hasil

  // BITWISE
  print("\nBitwise:");
  print(a & b); // AND bit
  print(a | b); // OR bit
  print(a ^ b); // XOR bit
  print(~a); // NOT bit
  print(a << 1); // geser bit kiri
  print(a >> 1); // geser bit kanan
 }
}
```

Output:

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```
Building package executable...
Built semua_operator:semua_operator.
Aritmatika:
13
7
30
3.3333333333333335
3
1

Increment Decrement:
5
7
7
5

Relasional:
false
true
true
false
true
false

Logika:
false
true
false

Assignment:
0

Null-aware:
Default
Budi
4
```

```
Type Test:
true
true
halo

Ternary:
Lulus

Bitwise:
2
11
9
-11
20
5
```

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